FlowPainter

Overview

`FlowPainter` is a Unity Editor tool for painting `FlowMap` textures directly on top of a source image.

Main features:

- Drag a source texture into the canvas and paint flow directions directly
- In `FlowMap` view, drag in an existing FlowMap and continue painting from it; mismatched sizes can be scaled to the source size
- Switch between Effect Preview, `FlowMap`, and Source view in real time
- Paint, erase, undo, redo, and reset the viewport
- Pan and zoom the canvas while painting
- Export a standard `FlowMap PNG`

Open the tool from:

- `Window/TLNEXUS/PicSuite/FlowPainter`
- Or select a texture in `Project` and use `Assets/TLNEXUS/Paint Flowmap`

Feature Notes

- Brush Size: Controls the brush area
- Brush Strength: Controls how strongly the current drag direction is written into the `FlowMap`
- Brush Softness: Controls how soft the brush edge is
- Brush Spacing: Controls the spacing between brush samples while dragging
- Preview Strength: Controls the flow intensity in the in-window animated preview
- Preview Speed: Controls the playback speed of the in-window animated preview
- Preview Tiled Sampling: Controls whether the preview uses Repeat or Clamp outside the texture bounds
- Export Texture Repeat Wrap: Controls the wrap mode import setting applied to the exported PNG
- Save to Original Folder: Exports directly to the source texture folder
- Reset FlowMap: Clears the current painted result back to the neutral state

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Workflow

1. Open `FlowPainter`
2. In Source view, drag a source texture into the canvas on the right
3. Use the top toolbar to switch between:

- Effect Preview
- `FlowMap`
- Source
- Paint / Erase

1. Adjust the brush and preview settings on the left

2. Optional: switch to `FlowMap` view and drag in an existing FlowMap; confirm `Scale and Load` when its dimensions differ from the source texture
3. Drag on the canvas to paint flow directions
4. Check the preview result
5. Click `Export FlowMap`

Shortcuts

- Left drag: Paint flow direction
- `Shift + Left drag horizontally`: Adjust brush size
- `[` / `]`: Decrease / increase brush size
- `Ctrl + Left drag horizontally`: Adjust brush strength
- Middle mouse drag: Pan canvas
- Mouse wheel: Zoom canvas

Other Buttons

- Undo: Revert the previous paint step
- Redo: Restore the reverted paint step
- Reset View: Restore the canvas to its default centered view
- Reset FlowMap: Clear the current painted result
- Save to Original Folder: Export into the same folder as the source texture

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Export Notes

- Default export name: `SourceTextureName_FlowMap.png`
- Preview settings affect only the in-window preview and do not modify exported pixel data
- `Export Texture Repeat Wrap` affects the import setting of the exported PNG
- Loading an existing FlowMap copies its pixels into the working canvas; the original asset is not modified
- When dimensions differ, `Scale and Load` bilinearly resamples the FlowMap to the source texture dimensions; cancel leaves the current canvas unchanged